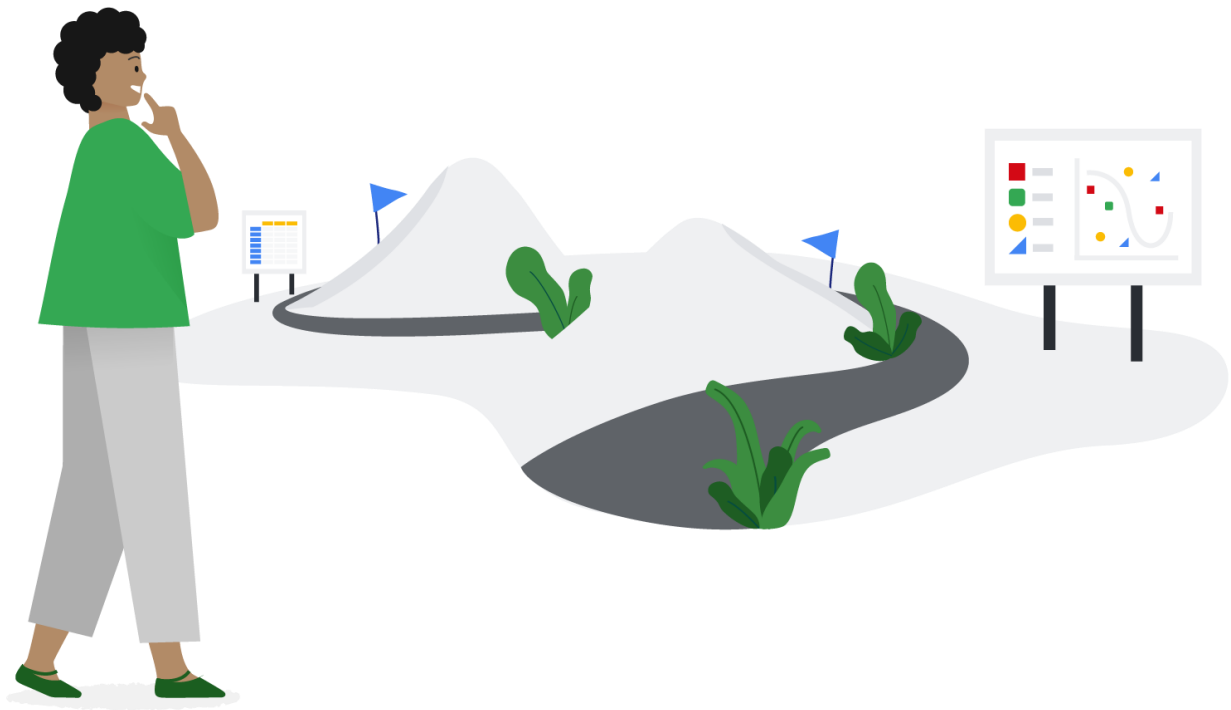




Roadmap Refresher: Your Google Data Analytics Certificate

Your Google Data Analytics Certificate roadmap

Use this guide to review the topics covered, tools used, and skills you will use in each course.



Which course?

For details on topics covered, tools used, and skills you will build, select the course you'd like to learn more about.

Select a course

1. Foundations
2. Ask
3. Prepare
4. Process
5. Analyze
6. Share
7. Act
8. Capstone

Frequently asked questions

What tools or platforms are included in the curriculum?

Spreadsheets, SQL, presentation tools, Tableau, RStudio, and Kaggle.

Do I need to take the course in a certain order?

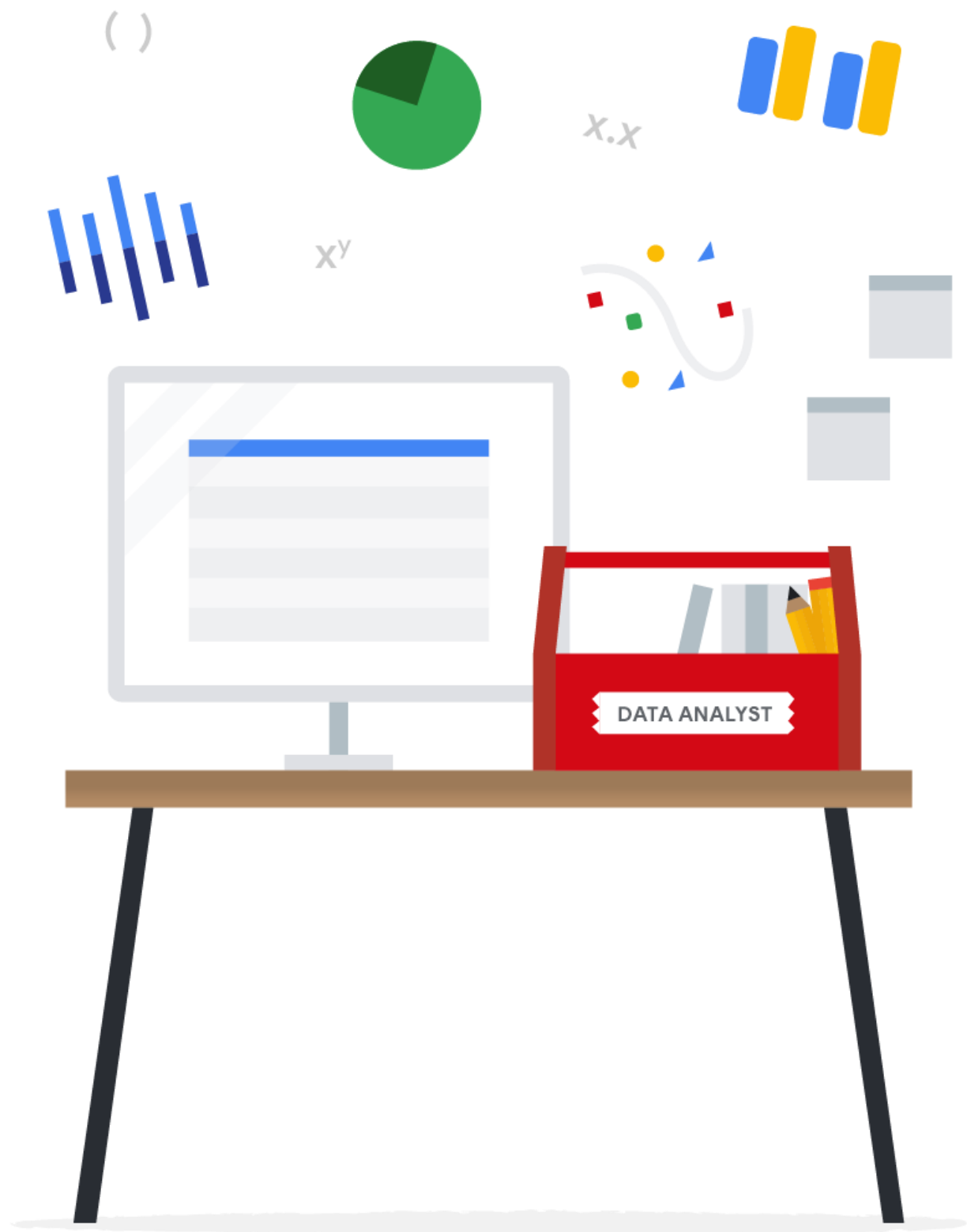
We highly recommend completing the courses in the order presented because the content in each course builds on information from earlier lessons.

Will you be teaching R or Python?

This program teaches the open-source programming language, R, which is great for foundational data analysis, and offers helpful packages for beginners to apply to their projects. We do not cover Python in the curriculum.

Course 1:
Foundations:

Course 1:
Foundations:



What you will learn:

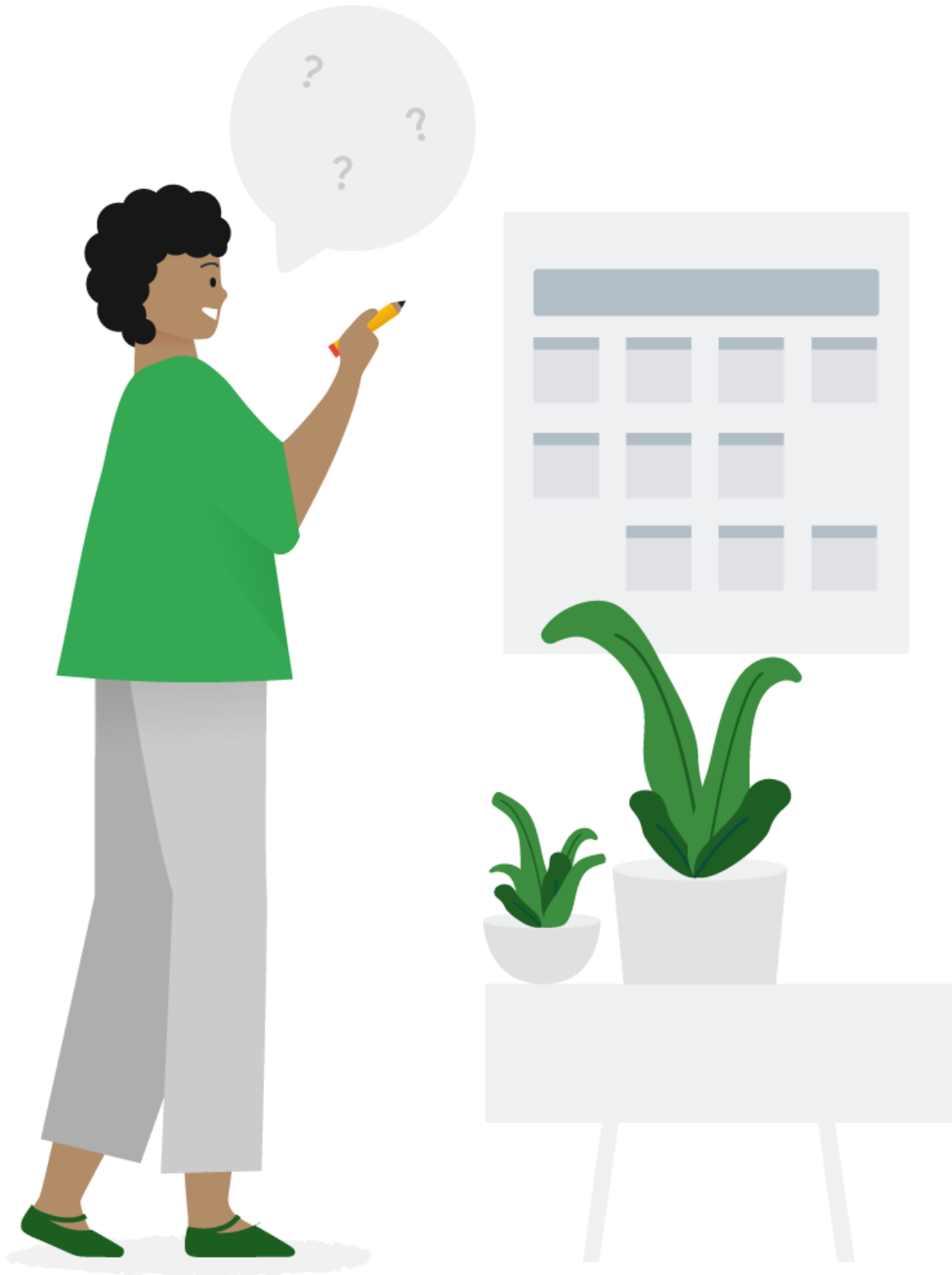
- Real-life roles and responsibilities of a junior data analyst
- How businesses transform data into actionable insights
- Spreadsheet basics
- Database and query basics
- Data visualization basics

Skill sets you will build:

- Using data in everyday life
- Thinking analytically
- Applying tools from the data analytics toolkit
- Showing trends and patterns with data visualizations
- Ensuring your data analysis is fair

Course 2:

Ask:



What you will learn:

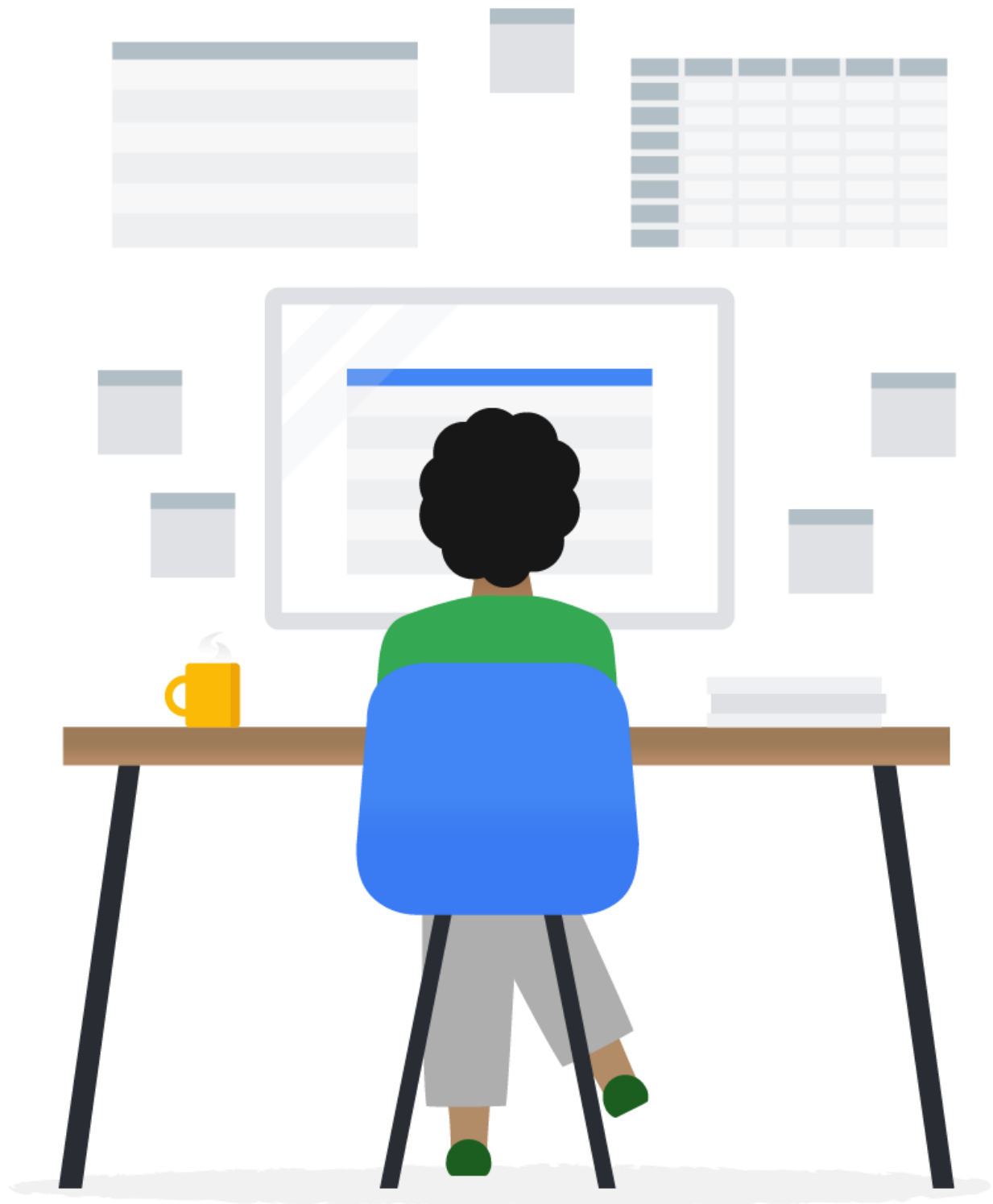
- How data analysts solve problems with data
- The use of analytics for making data-driven decisions
- Spreadsheet formulas and functions
- Dashboard basics, including an introduction to Tableau
- Data reporting basics

Skill sets you will build:

- Asking SMART and effective questions
- Structuring how you think
- Summarizing data
- Putting things into context
- Managing team and stakeholder expectations
- Problem-solving and conflict-resolution

Course 3:

Prepare:



What you will learn:

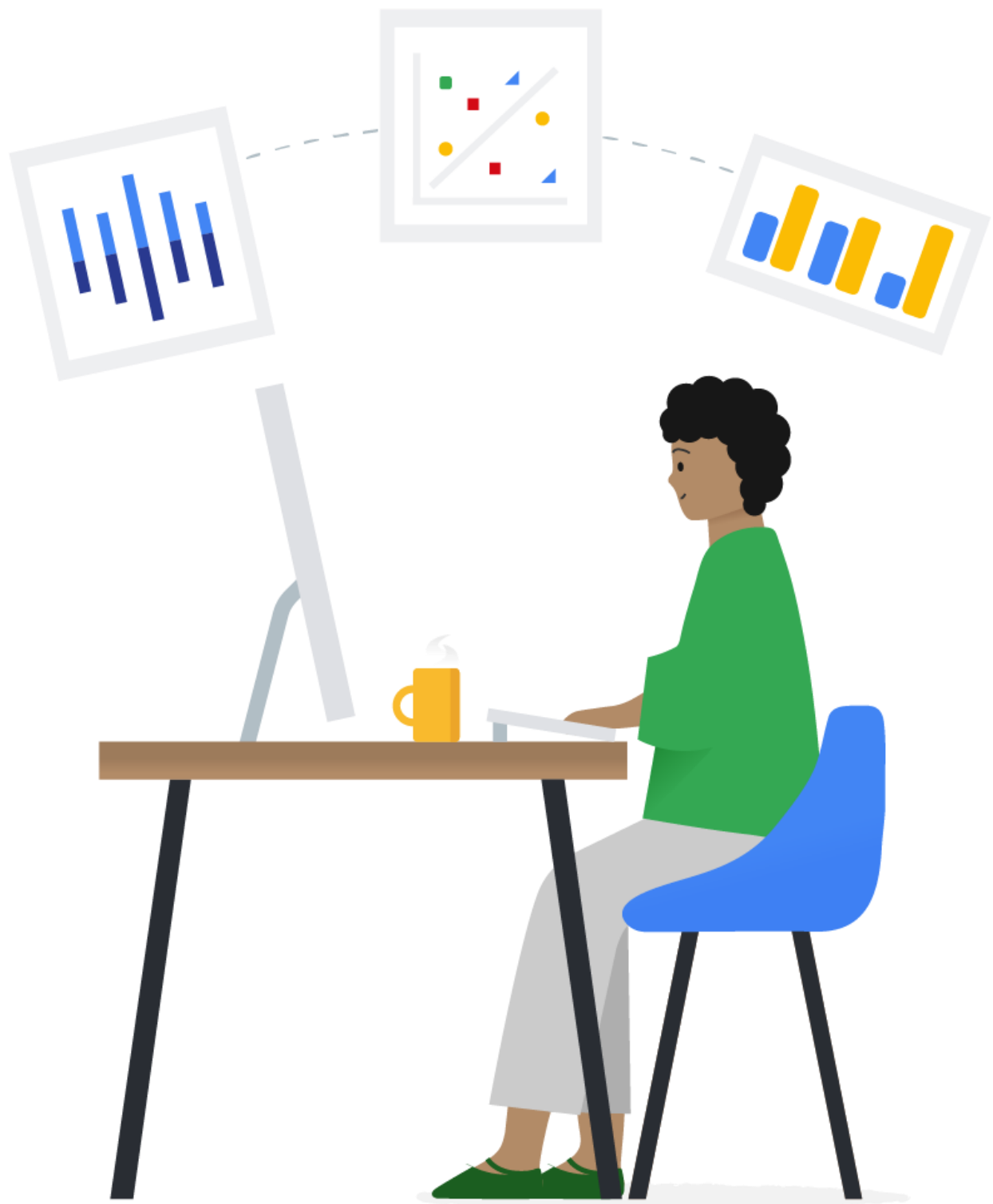
- How data is generated
- Features of different data types, fields, and values
- Database structures
- The function of metadata in data analytics
- Structured Query Language (SQL) functions

Skill sets you will build:

- Ensuring ethical data analysis practices
- Addressing issues of bias and credibility
- Accessing databases and importing data
- Writing simple queries
- Organizing and protecting data

Course 4:

Process:



What you will learn:

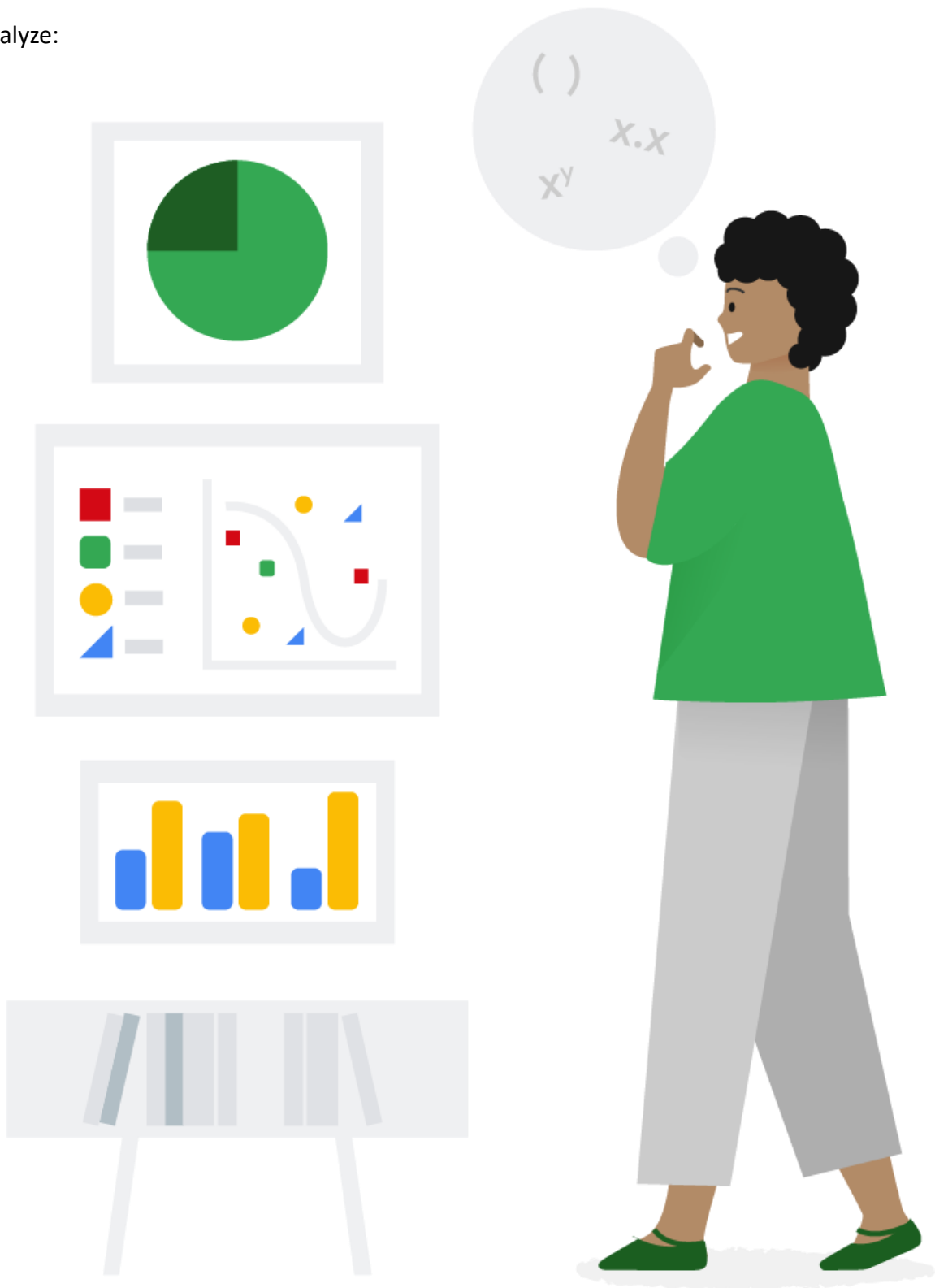
- Data integrity and the importance of clean data
- The tools and processes used by data analysts to clean data
- Data-cleaning verification and reports
- Statistics, hypothesis testing, and margin of error
- Resume building and interpretation of job postings (optional)

Skill sets you will build:

- Connecting business objectives to data analysis
- Identifying clean and dirty data
- Cleaning small datasets using spreadsheet tools
- Cleaning large datasets by writing SQL queries
- Documenting data-cleaning processes

Course 5:

Analyze:



What you will learn:

- Steps data analysts take to organize data
- How to combine data from multiple sources
- Spreadsheet calculations and pivot tables
- SQL calculations
- Temporary tables
- Data validation

Skill sets you will build:

- Sorting data in spreadsheets and by writing SQL queries
- Filtering data in spreadsheets and by writing SQL queries
- Converting data
- Formatting data
- Substantiating data analysis processes

Course 6:

Share:



What you will learn:

- Design thinking
- How data analysts use visualizations to communicate about data
- The benefits of Tableau for presenting data analysis findings
- Data-driven storytelling
- Dashboards and dashboard filters
- Strategies for creating an effective data presentation

Skill sets you will build:

- Creating visualizations and dashboards in Tableau
- Addressing accessibility issues when communicating about data
- Understanding the purpose of different business communication tools
- Telling a data-driven story
- Presenting to others about data

Course 7:

Act:



What you will learn:

- Programming languages and environments
- R packages
- R functions, variables, data types, pipes, and vectors
- R data frames
- Bias and credibility in R
- R visualization tools
- R Markdown for documentation, creating structure, and emphasis

Skill sets you will build:

- Coding in R
- Writing functions in R
- Accessing data in R
- Cleaning data in R
- Generating data visualizations in R

Course 8:

Capstone:



What you will learn:

- How a data analytics portfolio distinguishes you from other candidates
- Practical, real-world problem-solving
- Strategies for extracting insights from data
- Clear presentation of data findings
- Motivation and ability to take initiative

Skill sets you will build:

- Building a portfolio
- Increasing your employability
- Showcasing your data analytics knowledge, skill, and technical expertise
- Sharing your work during an interview
- Communicating your unique value proposition to a potential employer

You're done!

You've reviewed the tools and skill sets for each course in Google Data Analytics Certificate. Feel free to return to this activity for reference.

